

Lakshay Mittal

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Summary

Software Engineer specializing in backend and distributed systems, building secure, scalable services with Java, Spring Boot, and cloud-native tooling. Experienced in data pipeline orchestration, container-based deployment, and applying LLM and agentic AI workflows to engineering problems. Solid algorithms and data structures background; experienced in code review and cross-team collaboration.

Skills

Languages - Java, JavaScript/TypeScript, Python, SQL, HTML/CSS

Cloud & Databases - AWS (EC2, Lambda, S3, DynamoDB), Azure (Cosmos DB, Azure Functions, AKS), PostgreSQL, MongoDB, Redis

Libraries & Frameworks - ReactJS, Redux, Tailwind, NodeJS, ExpressJS, Spring Boot, Spring Security, Spring Data JPA (Hibernate)

Tools & Technologies - Kafka, Maven, Docker, Kubernetes, Swagger, JUnit, Mockito, Jest, Airflow, Claude Code, Spring AI, Datadog

Experience

Software Engineer | TechMentee, Inc.

Feb 2026 - Present | Alpharetta, GA (Remote)

- Engineered a secure service-to-service gateway in Java and Spring Boot with JWT-based authorization and OpenAPI schema validation across 10+ internal APIs.
- Diagnosed and eliminated N+1 query patterns in Spring Data JPA repositories using entity graphs and batch fetching, reducing redundant database round trips on high-traffic endpoints.
- Orchestrated Apache Airflow pipelines with idempotent tasks and automated retries, adding LLM-generated failure summaries that cut manual triage time by 40%.
- Containerized services with Docker and deployed to Kubernetes using rolling updates, readiness probes, and preStop graceful shutdown, enabling zero-downtime releases across environments.
- Instrumented services with Datadog dashboards and alerting, driving on-call incident response and root-cause analysis to reduce mean time to detection.

Software Engineer / Research Assistant | San Francisco State University

Jan 2025 - Feb 2026 | San Francisco, CA

- Decomposed core business logic into Java and Spring Boot microservices, improving scalability and maintainability.
- Normalized a PostgreSQL schema to eliminate duplicate records, offsetting the added join cost with composite and covering indexes to keep read latency flat.
- Architected shared state management using Redux and custom React hooks, handling complex user workflows while improving component reusability and reducing code duplication.
- Raised Jest unit-test coverage on the React/Redux layer to 95%, including reducer and async-thunk paths that previously had no regression coverage.

Systems Engineer | Infosys

Sep 2021 - Jun 2022 | Chandigarh, India

- Migrated critical event-driven services to AWS Lambda, replacing always-on compute with auto-scaling handlers and cutting idle infrastructure cost.
- Developed RESTful APIs using Java and Spring Boot, improving service response times by 20%.
- Implemented distributed caching with AWS ElastiCache (Redis) to reduce database load, achieving a 15% improvement in response times during peak traffic.

Education

San Francisco State University

Aug 2022 - Jan 2025

Master of Science, Computer Science

GPA: 3.74/4.0

Panjab University

Aug 2017 - Jun 2021

Bachelor of Engineering, Computer Science & Engineering

GPA: 3.58/4.0

Projects

HealthShare: AI-Enhanced Pandemic Data Repository | Java, PostgreSQL, ReactJS, AWS Lambda, ExpressJS, Llama, D3JS

- Developed an analytics platform containing 120K+ entries of healthcare professionals' experiences during COVID-19.
- Wrote Spring Cloud Function handlers on AWS Lambda for async filtering and batch transforms, with an Express API serving the React dashboard; integrated Llama language model for thematic classification of free-text responses.
- Created an interactive visualization suite with D3JS, including drill-down dashboards and multi-dimensional query controls.

DocQuery: Grounded Document Q&A Service | Java, Spring Boot, Spring AI, PostgreSQL/pgvector, Docker

- Built a document Q&A service in Spring Boot with schema-validated model responses, per-call timeouts and bounded retries, and a retrieval-only fallback when cited spans failed to match retrieved chunks.
- Implemented hybrid retrieval over pgvector and Postgres full-text search, fusing candidates with Reciprocal Rank Fusion behind a custom Spring AI DocumentRetriever.
- Designed chunk-level ingestion with content-hash deduplication, re-embedding only changed chunks on re-ingest.